Wearable Ear-Centric Physiological Sensing With On-Edge Physics-Informed Neural Networks for Negative Mental State Detection

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***Abstract*—Monitoring mental states is essential for understanding physiological and neurological health, offering insights that enhance diagnostics and therapy. However, current systems are often bulky, immobile, and reliant on external devices for deep learning (DL), limiting real-time applications. This study introduces a compact, real-time mental state monitoring system with four innovations. First, a wearable ear-centric device captures multimodal biosignals-behind-the-ear (BTE) electroencephalography (EEG) and in-ear photoplethysmography (PPG)-enhancing portability. Second, a high-resolution superlet transform (HRST) converts these signals into spectrogram images to enrich input features for DL models. Third, the PhysAttenConvLSTM model, designed for edge deployment using tiny machine learning (TinyML), integrates convolutional neural networks (CNNs), long short-term memory (LSTM), self-attention, and a physics-informed merge layer (PIML) to classify affective states efficiently. All processes—from signal acquisition to classification—occur directly on the edge device, enabling low-latency, real-time operation without external computation. Fourth, the Internet of Medical Things (IoMT) framework is implemented for data storage, remote access, and system management. Multimodal data were collected from 25 participants during social media-based mental state induction tasks. The system achieved 92.07% accuracy in leave-one-out cross-validation (LOOCV) and 95.05% in ten-fold cross-validation (CV), with strong specificity, sensitivity, precision, and *F*1 scores. This research advances edge DL, multimodal biosignal fusion, and IoMT integration, offering an ultraportable, energy-efficient solution for real-time mental state monitoring, with potential in affective computing and personalized mental healthcare.**

***Index Terms*—Behind-the-ear (BTE) electroencephalography (EEG), in-ear photoplethysmography (PPG), Internet of Medical Things (IoMT), mental state monitoring, on-edge computing, physics-informed neural networks (PINNs), tiny machine learning (TinyML).**

I. Introduction

MENTAL health disorders, such as anxiety, chronic stress, and depression, are growing global concerns, affecting millions and placing significant strain on healthcare systems [1], [2], [3]. Despite their prevalence, these conditions often remain undetected or inadequately treated, largely due to the lack of accessible, objective monitoring tools outside clinical settings. Early and reliable detection is vital for timely intervention and better outcomes. Traditional assessments—relying on self-reports and structured questionnaires, such as the Patient Health Questionnaire-9 (PHQ-9) [4], Generalized Anxiety Disorder-7 (GAD-7) [5], and Perceived Stress Scale (PSS) [6]—are constrained by subjectivity, recall bias, and infrequent measurement, limiting their ability to capture real-time emotional fluctuations in everyday contexts [7], [8]. In response, there is increasing interest in continuous, noninvasive monitoring systems that leverage wearable physiological sensors and embedded artificial intelligence [2], [7], [9]. These technologies can collect and interpret biosignals, such as electroencephalography (EEG), photoplethysmography (PPG), electrodermal activity (EDA), heart rate variability (HRV), and respiration, which reflect underlying neural and autonomic states associated with mental distress [10], [11], [12]. By enabling real-time, context-aware tracking of negative emotional states, these systems offer the potential to transform mental healthcare. Rather than relying solely on clinic-based models, they support a shift toward personalized, preventive approaches that operate seamlessly in daily life, offering new possibilities for managing and mitigating the impacts of mental health disorders.

Recent advancements in physiological sensing technologies and embedded intelligence have paved the way for wearable systems capable of continuous, real-time mental health monitoring in naturalistic settings [2], [7], [13]. These systems aim to move beyond sporadic clinical assessments by enabling unobtrusive tracking of mental state fluctuations throughout daily life. Among the wide array of biosignals, EEG and PPG have emerged as two highly informative and complementary modalities [11], [12], [14], [15]. EEG measures the brain's electrical activity with millisecond precision, providing insights into cognitive load, emotional arousal, and stress-related neural oscillations [16]. PPG quantifies blood volume changes induced by cardiac cycles, offering indirect but robust indicators of autonomic nervous system (ANS) regulation, such as HRV and pulse amplitude variability—both of which are highly sensitive to emotional and physiological stress [16], [17].

Various configurations of these sensing techniques have been explored. Scalp-based EEG systems provide high signal quality and comprehensive spatial coverage, making them the gold standard for research-grade neural recordings and

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clinical diagnostics [18]. Wrist- and finger-based PPG devices, widely available in commercial fitness trackers and smartwatches, offer ease of use, low cost, and effective long-term monitoring of cardiovascular trends [19]. Similarly, functional near-infrared spectroscopy (fNIRS) and EDA have demonstrated utility in measuring cortical hemodynamics and sweat gland activity, respectively—both relevant in stress and emotion studies [20], [21]. These approaches have contributed significantly to the development of mental state monitoring systems.

However, these traditional configurations have notable limitations for continuous, real-world deployment. Scalp EEG is bulky and intrusive, often requiring conductive gels and extensive setup, while wrist- and finger-based PPG is prone to motion artifacts and may lose accuracy during dynamic activities [19], [22]. EDA and fNIRS systems are often sensitive to ambient conditions or require complex calibration [21], [23]. In contrast, ear-centric physiological sensing, particularly using behind-the-ear (BTE) EEG and in-ear PPG, offers a promising compromise between signal quality, wearability, and usability [22], [24].

The proximity of the ear region to both cerebral and major vascular structures (e.g., the temporal lobe and superficial temporal artery) allows for simultaneous acquisition of high-fidelity neural and cardiovascular signals [18], [22]. BTE EEG captures stress-sensitive brain activity—such as changes in alpha and beta rhythms—with greater user mobility and comfort compared to scalp systems [18], [22], [25]. In-ear PPG sensors, embedded discreetly within custom earpieces, benefit from a stable optical path less affected by motion, delivering reliable data on HRV and pulse dynamics even during movement [24], [26]. Additionally, the anatomical consistency and semi-rigid structure of the ear canal allow for reproducible sensor placement and better skin contact, which enhances signal stability. By combining these modalities, ear-centric sensing enables a multimodal and minimally obtrusive solution for detecting negative mental states, such as stress, anxiety, or cognitive overload. This approach bridges the gap between clinical-grade monitoring and user-friendly wearability, forming the foundation for next-generation real-time mental health assessment tools.

Despite significant progress in wearable mental health technologies, many existing systems still depend heavily on external computing resources—such as smartphones, PCs, cloud servers, or edge gateways—for signal preprocessing, feature extraction, and classification [18], [22]. While this architecture leverages powerful computational capabilities, it introduces critical limitations, including increased latency, higher energy consumption, and reliance on stable connectivity. These issues are particularly detrimental to continuous, real-time mental state monitoring in mobile or uncontrolled environments, where low-latency response, long battery life, and user privacy are essential.

To overcome these challenges, on-edge processing—where all computational stages are executed locally on the wearable device—has emerged as a promising solution. Tiny machine learning (TinyML) supports the deployment of optimized machine learning models on low-power microcontroller units (MCUs) and IoT edge devices [27]. This enables real-time inference directly on the device, reducing power usage and removing the need for constant wireless communication. Additionally, TinyML enhances privacy by keeping sensitive biosignals, such as EEG and PPG entirely on-device.

However, most deep learning (DL) models used in TinyML treat biosignals as black-box inputs, often ignoring physiological knowledge or signal-generation mechanisms. This can result in overfitting, limited generalizability, and poor interpretability—especially in small-sample or subject-variable wearable health settings [22]. Conventional architectures, such as convolutional neural networks (CNNs) or long short-term memories (LSTMs), may capture spurious or nongeneralizable features. To address this, there is growing interest in physics-informed neural networks (PINNs) and hybrid models that integrate physiological priors into the learning process [28]. When implemented in TinyML-compatible frameworks, these physics-informed approaches offer a compelling path toward energy-efficient, interpretable, and real-time detection of mental distress on wearable, ear-centric platforms [29].

In this study, we introduce a novel ear-centric wearable physiological sensing system that integrates BTE EEG and in-ear PPG with PINNs, fully deployed on-edge for real-time detection of negative mental states. By embedding domain-specific knowledge—such as characteristic brainwave rhythms and cardiovascular pulse dynamics—into the neural network architecture, the system enhances both model robustness and interpretability while maintaining the energy efficiency advantages of edge-based processing. Operating continuously without entering sleep mode, the device consumes approximately 85.193 mW and achieves an operational duration of around 21.71 h on a single charge. As presented in Section III, this energy-efficient performance substantially surpasses that of the existing biosignal monitoring devices. Furthermore, the system eliminates reliance on external computing resources, thereby minimizing communication latency and reducing overall power consumption.

The key contributions of this article include the following.

1) The design of a compact, dual-modality ear-centric sensing device that simultaneously acquires BTE EEG and in-ear PPG signals for high-fidelity, multimodal physiological monitoring.
2) The development of a PINN architecture optimized for embedded deployment, incorporating temporal and spectral priors from EEG and PPG to enhance mental state assessment accuracy.
3) The implementation of an efficient on-edge pipeline that performs real-time signal transformation, feature extraction, and classification of negative mental states, with seamless integration of the Internet of Medical Things (IoMT) capabilities for remote monitoring.
4) A comprehensive evaluation demonstrating superior energy efficiency, responsiveness, and detection accuracy compared to the existing wearable systems that rely on external computing infrastructure.

The structure of this article is organized as follows. Section II outlines the proposed framework, the experimental procedures, and its underlying methodology. Section III eval-

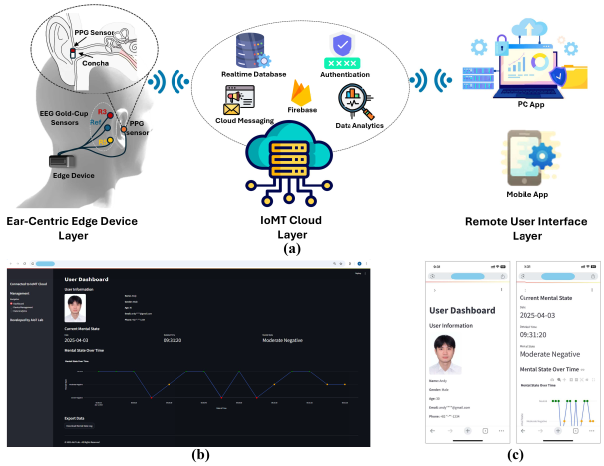

Fig. 1. (a) Overview of the core components comprising the IoMT-based mental state monitoring system, (b) web-based platform for real-time mental state tracking, and (c) mobile application for user interaction and data access.

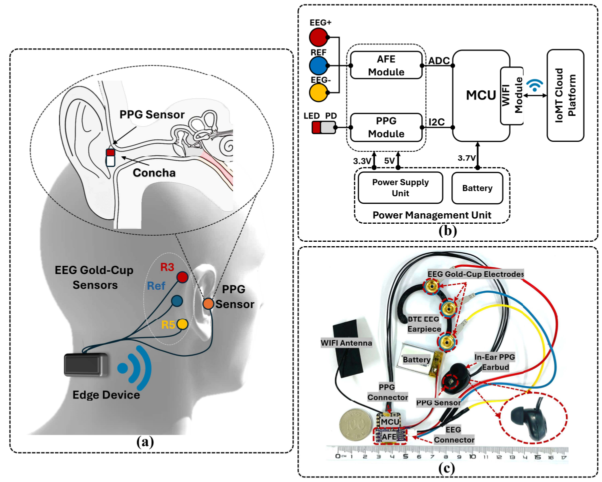

Fig. 2. (a) Measurement locations and multimodal ear-centric edge device, (b) hardware schematic, and (c) assembled system.

uates the system's performance. Finally, Section IV concludes the study and highlights potential avenues for future research.

II. Materials and Methods

A. Framework Description

This study introduces a comprehensive framework for the early and remote detection of negative mental states, leveraging ear-centric physiological signals, edge computing capabilities, and IoMT technologies. As depicted in Fig. 1, the system is structured into three core components.

1) Ear-Centric Edge Device Layer: The ear-centric wearable edge device is developed to acquire multimodal biosignals, specifically EEG signals from BTE and PPG signals from within the ear canal, thereby enhancing portability and overall user experience. All signal processing, feature transformation, and classification are performed locally on the device, enabling real-time operation with minimal latency and eliminating the need for external computational resources. Integrated Wi-Fi connectivity supports seamless communication with an IoMT cloud platform, facilitating synchronized data management and enabling advanced remote analytics.

2) IoMT Cloud Server Layer: The processed data from the edge device are transmitted to an IoMT cloud server hosted on the Firebase platform, enabling real-time data synchronization, secure storage, and advanced analytics for the continuous monitoring of negative mental states based on multimodal biosignals. This cloud-based infrastructure supports efficient data management and visualization, thereby facilitating ongoing mental state tracking and comprehensive analysis.

3) IoMT Remote User Interface Layer: Remote monitoring applications—accessible via both PC-based and mobile platforms (shown in Fig. 1)—have been developed using an integration of Streamlit and Firebase to enable real-time visualization of users' mental state levels by both individuals and healthcare professionals. These applications feature alert notification systems that respond to the detection of negative mental states, thereby supporting timely and appropriate interventions. By leveraging ear-centric physiological signals, edge computing, and IoMT technologies, the system offers a robust and continuous monitoring framework for the detection and management of adverse mental health conditions. Through these applications, authorized healthcare providers and stakeholders can access data from the IoMT cloud server, facilitating remote supervision and prompt clinical response. This functionality ensures that medical personnel can act swiftly and effectively when intervention is required.

B. Multimodal Ear-Centric Edge Device

The proposed edge-computing earpiece is engineered for the concurrent acquisition of EEG signals from the area BTE and PPG signals from within the ear canal. With a compact form factor measuring 2 $\times$ 2.5 cm, the device ensures both user comfort and wearability overextended periods (see Fig. 2). EEG data are collected using three gold-cup electrodes: the blue electrode serves as the reference, while the red and yellow electrodes function as active biopotential sensors. These are positioned according to the cEEGrid configuration at R3, R5, and the reference site, as illustrated in Fig. 2(a) [30]. The EEG signals are amplified via the BioAmp EXG Pill analog front end and digitized by the MCUs analog-to-digital converter (ADC) [31]. For PPG signal acquisition, a MAX30102 pulse oximeter module is embedded at the concha location inside the earbud [Fig. 2(a)], a site validated in prior studies for generating high-quality PPG signals [24], [26]. This module integrates red and infrared LEDs, photodetectors with spectral sensitivity from 640 to 980 nm, and built-in optical and ambient light rejection to enhance signal fidelity. Digitized PPG data are transmitted to the MCU via the I^2C interface. The digitized PPG signals are communicated to the MCU via the I^2C protocol. Both EEG and PPG signals are sampled at 125 Hz to ensure synchronized acquisition. Signal processing, feature transformation, and classification are performed entirely on-board using the ESP32, a microcontroller featuring a 32-bit RISC-V CPU with an integrated floating-point unit (FPU), allowing for efficient, real-time processing without the need for external computational resources. Only the final

Algorithm 1 Labeling Procedure

Input:
- $DAS_{score} \in \{0, 1, 2, 3\}$ (DAS self-assessment)
- β_s: EEG beta power 13–30 Hz
- HRV_s: PPG heart rate variability
- μ_β, σ_β: baseline EEG beta mean and standard deviation
- μ_{HRV}, σ_{HRV}: baseline HRV mean and standard deviation

Output: MentalStateLabel: $Label_{final}$
- $Label_{final} \in \{Neutral, Moderate\ Negative, Severe\ Negative\}$

Step 1: DAS-Based Labeling

1. **If** $DAS_{score} = 0$, **set** $Label_{DAS} \longleftarrow Neutral$
2. **Else if** $DAS_{score} \in \{1, 2\}$, **set** $Label_{DAS} \longleftarrow Moderate\ Negative$
3. **Else if** $DAS_{score} = 3$, **set** $Label_{DAS} \longleftarrow Severe\ Negative$

Step 2: EEG Beta-Based Labeling

1. **Compute dynamic thresholds:** $T^{\beta}_{low} = \mu_\beta - 0.5\sigma_\beta, T^{\beta}_{high} = \mu_\beta + 1.5\sigma_\beta$
2. **If** $\beta_s < T^{\beta}_{low}$, **set** $Label_{EEG} \longleftarrow Neutral$
3. **Else if** $T^{\beta}_{low} < \beta_s < T^{\beta}_{high}$, **set** $Label_{EEG} \longleftarrow Moderate\ Negative$
4. **Else, set** $Label_{EEG} \longleftarrow Severe\ Negative$

Step 3: PPG HRV-Based Labeling

1. **Compute dynamic thresholds:** $T^{HRV}_{low} = \mu_{HRV} - 0.5\sigma_{HRV}, T^{HRV}_{high} = \mu_{HRV} + 1.5\sigma_{HRV}$
2. **If** $HRV_s > T^{HRV}_{high}$, **set** $Label_{PPG} \longleftarrow Neutral$
3. **Else if** $T^{HRV}_{low} < HRV_s < T^{HRV}_{high}$, **set** $Label_{PPG} \longleftarrow Moderate\ Negative$
4. **Else, set** $Label_{PPG} \longleftarrow Severe\ Negative$

Step 4: Final Label Fusion

1. **If** $Label_{DAS} = Label_{EEG} = Label_{PPG}$, **then** $Label_{final} \longleftarrow Label_{DAS}$
2. **Else if** $|\beta_s - \mu_\beta| > 2\sigma_\beta$ or $|HRV_s - \mu_{HRV}| > 2\sigma_{HRV}$ **then** $Label_{final} \longleftarrow Label_{EEGorPPG}$ (with the largest deviation)
3. **Else** $Label_{final} \longleftarrow Label_{DAS}$
4. **Return** $Label_{final}$

processed results are transmitted to a cloud-based IoT platform via Wi-Fi for further analysis. Power is supplied by a 3.7-V, 500-mAh lithium battery, regulated by an integrated power management system. The power supply unit includes the MCP73831 battery charge management controller, ensuring safe and efficient charging and energy delivery to all system components, thereby supporting reliable, continuous physiological monitoring.

C. Experimental Protocol

Twenty-five students (17 males and 8 females) from Pukyong National University participated in this study. All participants were in good health and reported no history of visual impairments or sleep-related disorders. The cohort consisted of both undergraduate and graduate students from diverse national backgrounds, including South Korea, Vietnam, India, Pakistan, Bangladesh, Indonesia, and Iran. Detailed demographic information is provided in Table I. The study aimed to detect negative mental states induced during social media use by employing a wearable, ear-centered physiological monitoring device that concurrently recorded EEG and PPG signals at a sampling rate of 125 Hz. To control for potential confounding variables, participants were instructed to abstain from alcohol and stimulants prior to the experiment. The protocol simulated real-world social media usage through three activity types: passive content browsing (e.g., Instagram, Facebook), interaction with emotionally provocative comments (e.g., X, Reddit), and viewing emotionally charged short-form videos (e.g., TikTok, YouTube Shorts) covering topics, such as body image, social injustice, and social comparison. The types of social media content were pre-evaluated and categorized into three emotional intensity levels: "neutral," "moderate negative," and "severe negative." The specific content presented in each trial was randomly selected from the prelabeled pool to avoid bias. Each participant completed five trials per activity across three experimental sessions. As illustrated in Fig. 3, each session comprised four phases: a 5-s task cue, 5 min of content engagement, a 30-s self-assessment, and a 5-min resting period for recovery. Following each trial, participants completed a depression, anxiety, and stress (DAS) self-assessment using a four-point Likert scale [32], [33]. DAS scores were categorized into three mental states: neutral (0), moderate negative (1)–(2), and severe negative (3), and were used to annotate the corresponding EEG and PPG data segments. To enhance objectivity and temporal resolution, physiological markers were extracted from both EEG and PPG signals: EEG beta-band power (13–30 Hz) and PPG-derived HRV [3], [8], [34]. Individualized baselines were established for each participant using pretask resting-state segments. Dynamic thresholds were applied to categorize physiological activity: values below ($\mu - 0.5\sigma$) were labeled as "neutral," those between ($\mu - 0.5\sigma$) and ($\mu + 1.5\sigma$) as "moderate negative," and those above ($\mu + 1.5\sigma$) as "severe negative," where μ and σ represent the mean and standard deviation of baseline beta power or HRV, respectively. This physiological annotation was motivated by prior findings that associate elevated EEG beta activity and reduced PPG-derived HRV with stress, arousal, and cognitive load [3], [8], [34]. For final label assignment, DAS, EEG, and PPG labels were fused using a rule-based strategy: if all labels agreed, that label was assigned; otherwise, the physiological label exhibiting the largest deviation from baseline (e.g., $|\beta - \mu| > 2\sigma$ or $|\text{HRV} - \mu| > 2\sigma$) was prioritized, and the DAS-based label was used when physiological deviations were marginal. This hybrid approach reduces subjectivity while maintaining high-confidence physiological grounding of mental state classification. The complete labeling process is outlined in Algorithm 1. This study was reviewed and approved by the Institutional Review Board of Pukyong National University (Approval No: 1041386-202305-HR-42-02).

D. High-Resolution Superlet Transform

The high-resolution superlet transform (HRST) is a powerful time–frequency analysis technique that enhances spectral

TABLE I
PARTICIPANT DEMOGRAPHICS

Demographic variables	Value
Age (years)	30.45 ± 6.27
Height (cm)	170.36 ± 7.29
BMI (kg/m^2)	23.73 ± 1.67
Sex (Male: Female)	17: 8

resolution by adaptively combining multiple Morlet wavelets with varying bandwidths [35]. This approach is particularly advantageous for converting physiological signals, such as PPG and EEG, into high-resolution spectrogram images for DL applications, including the detection of negative mental states. Given an input signal $x(t)$, the HRST computes time–frequency representations by leveraging a set of Morlet wavelets $\psi_{f,s}(t)$, where the wavelet scale s is defined as

$$s_n = \frac{1}{2\pi f \gamma_n} \tag{1}$$

and γ_n represents the bandwidth factor. The superlet transform is obtained by averaging multiple wavelets as

$$\Psi_f^{(N)}(t) = \frac{1}{N}\sum_{1}^{N} \psi_{f,s_n}(t) \tag{2}$$

which adaptively improves spectral resolution. In the HRST, the number of wavelets $N(f)$ varies with frequency, ensuring that lower frequencies receive higher resolution through increased wavelet superposition, while higher frequencies maintain temporal precision

$$N(f) = \left\lfloor N_{\max} \cdot \left(\frac{f - f_{\min}}{f_{\max} - f_{\min}}\right)\right\rfloor + 1. \tag{3}$$

Unlike conventional methods, such as short-time Fourier transform (STFT), which suffers from a fixed tradeoff between time and frequency resolution, HRST dynamically adjusts the number of superimposed wavelets, significantly enhancing spectral resolution without sacrificing temporal precision [35]. Similarly, compared to the continuous wavelet transform (CWT), which applies a single wavelet per frequency, HRST provides superior multiscale analysis by adaptively stacking multiple wavelets, yielding sharper frequency localization and minimizing spectral leakage. This is particularly beneficial for PPG signals, where subtle variations in HRV and vascular responses are crucial for detecting mental states, such as stress and relaxation. For EEG signals, HRST effectively distinguishes between overlapping frequency bands, enabling more precise detection of brain rhythms associated with emotional processing, such as increased frontal theta power during stress or elevated alpha power in relaxed states. The resulting HRST spectrogram is given by

$$\mathrm{HRST}(f,t) = \int x(t')\, \Psi_f^{(N(f))*}(t')\, dt'. \tag{4}$$

Furthermore, unlike empirical mode decomposition (EMD) [36] and the Hilbert–Huang transform (HHT) [37], which often introduce mode-mixing artifacts and require complex postprocessing, HRST offers a robust and computationally efficient alternative for mental state recognition. By converting PPG and EEG signals into high-resolution spectrogram images, HRST enables DL models to extract richer features, resulting in higher classification accuracy and more reliable mental state detection systems.

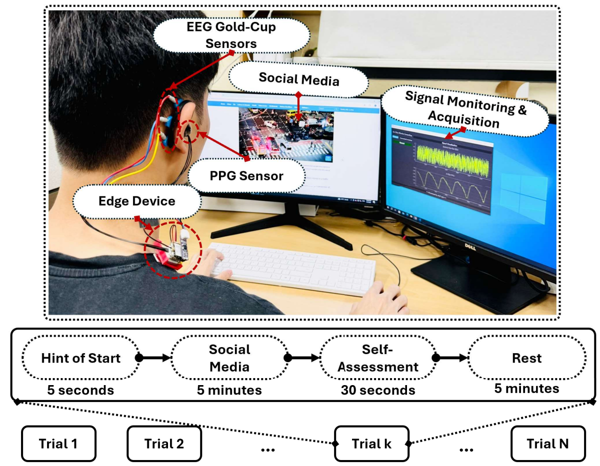

Fig. 3. Experimental setup.

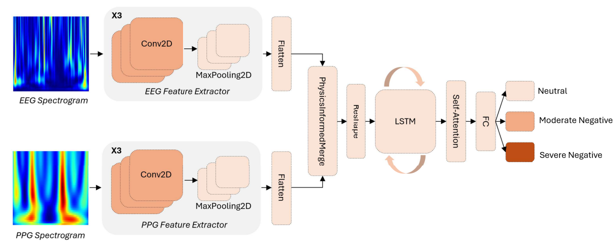

Fig. 4. Proposed PhysiAttenConvLSTM architecture for negative mental state classification.

E. Proposed PhysiAttenConvLSTM for Mental State Classification

The proposed model, PhysiAttenConvLSTM, integrates a CNN [38], LSTM [39], and a self-attention mechanism [40], along with a novel physics-informed merge layer [28], [29], to classify mental states using HRST-based EEG and PPG spectrogram inputs. The architecture is illustrated in Fig. 4 and detailed in Table II.

1) CNN-Based Feature Extractors: The CNN component serves as the feature extractor, learning spatial features from the raw EEG and PPG spectrogram images, which encode frequency-domain characteristics of the signals. The convolution operation is formulated as

$$y_{ij} = (W * x)_{ij} + b \tag{5}$$

where x is the input spectrogram, W is the convolution kernel, and b is the bias term. This convolutional process enables the model to capture local patterns indicative of mental state shifts, such as changes in rhythm and frequency. It is followed by max-pooling layers that reduce spatial dimensions and emphasize the most salient features. The CNN component

TABLE II
ARCHITECTURAL SPECIFICATIONS OF THE PROPOSED PHYSIATTENCONVLSTM MODEL

Hyper-parameters/ Layer	Search Space	Optimal Value: First branch (BTE EEG)	Optimal Value: Second branch (In-Ear PPG)	Operation	Param#	*AF	Output Shape
Input	28x28x1	28x28x1	28x28x1	---	---	---	---
Conv2D	[32, 64, 128]	32	32	k=3, s=1, p=1	320*2	ReLU	(-1, 28, 28, 32) | (-1, 28, 28, 32)
MaxPooling2D	---	---	---	k=2, s=1	---	---	(-1, 14, 14, 32) | (-1, 14, 14, 32)
Conv2D	[32, 64, 128]	64	64	k=3, s=1, p=1	18,496*2	ReLU	(-1, 14, 14, 64) | (-1, 14, 14, 64)
MaxPooling2D	---	---	---	k=2, s=1	---	---	(-1, 7, 7, 64) | (-1, 7, 7, 64)
Conv2D	[32, 64, 128]	128	128	k=3, s=1, p=1	73,856*2	ReLU	(-1, 7, 7, 128) | (-1, 7, 7, 128)
MaxPooling2D	---	---	---	k=2, s=1	---	---	(-1, 3, 3, 128) | (-1, 3, 3, 128)
Flatten	---	---	---	---	---	---	(-1, 1152) | (-1, 1152)
FIML	---	---		---	147,586	---	(-1, 64)
Reshape	---	---		---	---	---	(-1, 1, 64)
LSTM	[64, 128, 256]	128		---	98,816	---	(-1, 1, 128)
Self-Attention	---	---		---	128	---	(-1, 128)
Dense	[64, 128, 256]	128		---	16,512	ReLU	(-1, 128)
Output	3	3		---	387	Softmax	(-1, 3)
Optimizer	[SGD, Adam]	Adam		---	---	---	---
Batch Size	[8, 16, 32]	16		---	---	---	---
Learning Rate	[0.01, 0.001, 0.0001]	0.0001		---	---	---	---
Epoch	250	250		---	---	---	---

*AF: activation function, k: kernel size, s: stride, p: padding

is essential for learning the spectral and temporal patterns typically associated with different mental states [38].

2) LSTM With Self-Attention: The self-attention-enhanced LSTM represents a cutting-edge methodology in sequence data processing, specifically designed to capture both temporal patterns and long-term dependencies with high efficiency [39], [40]. By integrating the advantages of the conventional LSTM network with the self-attention mechanism, this approach enhances the model's ability to focus on critical information within sequential data. The LSTM component plays a fundamental role in preserving long-range dependencies, which is particularly important for tracking the evolution of mental states over time. Unlike standard recurrent neural networks (RNNs), LSTMs effectively address the vanishing gradient issue, allowing the model to retain and utilize information across extended sequences, thereby improving predictive performance in tasks involving sequential dependencies. LSTM units consist of three primary gates—forget gate (f_t), input gate (i_t), and output gate (o_t), along with a cell state (c_t), which carries memory across time steps. These gates interact to regulate the flow of information into and out of the cell state. Specifically, the forget gate decides which information from the previous cell state is discarded, while the input gate determines what new information will be stored. The cell state is updated using the following equation:

$$c_t = f_t.c_{t-1} + i_t.\tilde{c}_t \tag{6}$$

where $\tilde{c}_t$ is the candidate cell state, computed based on the current input and previous hidden state. The output gate then generates the final hidden state h_t, which is used in subsequent layers or predictions

$$h_t = o_t.\tanh(c_t). \tag{7}$$

This mechanism enables the LSTM to more effectively model the temporal dependencies of mental states, allowing it to learn transitions over time between neutral, moderately negative, and severely negative states.

3) Self-Attention Mechanism: The self-attention mechanism enhances the model's ability to focus on the most relevant features in the temporal sequence [40]. In contrast to traditional attention mechanisms that operate on the entire sequence, the self-attention mechanism dynamically weighs the importance of each input element by considering the interactions within the sequence itself. This approach is particularly effective in identifying the key time steps that contribute most to accurate mental state classification. Mathematically, the self-attention mechanism computes an attention score for each element of the sequence by using a scaled dot-product attention, given by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V \tag{8}$$

where Q, K, and V are the query, key, and value matrices, respectively, derived from the input sequence. The query and key matrices capture the relationships between sequence elements, while the value matrix holds the information to be weighted and aggregated. The attention score is computed as the dot product of the query and key matrices, scaled by $\sqrt{d_k}$, followed by a softmax function to normalize the scores. This allows the model to assign higher attention to the most relevant time steps or features, which is particularly effective in identifying key moments that influence mental states. In our approach, the self-attention layer is strategically placed between the LSTM layer and the final output layer to refine the extraction of meaningful patterns from sequential data. The LSTM layer first captures temporal dependencies by producing a sequence of hidden states, where each state retains information up to its respective time step. These hidden states are subsequently processed by the self-attention mechanism, which assigns adaptive weights based on the contextual relevance of each state within the sequence. This dynamic weighting enables the model to focus on the most influential time steps, thereby enhancing the overall representation of temporal dynamics and improving performance in sequence-based learning tasks.

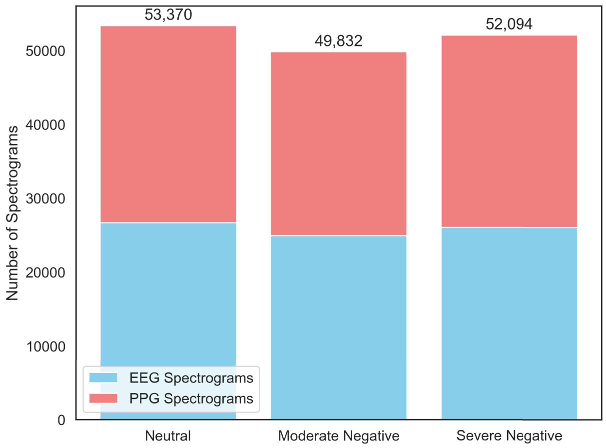

Fig. 5. Number of spectrogram samples after HRST for each class label: neutral, moderate negative, and severe negative.

4) Physics-Informed Merge Layer (PIML): The PIML is a novel component that effectively integrates the features extracted from the two CNN branches, which independently process the HRST-based EEG spectrogram and the HRST-based PPG spectrogram. This integration is accomplished through a weighted addition mechanism that allows the model to balance the contributions from both data streams. The fusion process is governed by the equation

$$\begin{aligned} y_k = \gamma_1 . f^E_{k-1}\left(W^E_{k-1} . y^E_{k-1}(|X^E(i)|) + b^E_{k-1}\right) \\ + \gamma_2 . f^P_{k-1}\left(W^P_{k-1} . y^P_{k-1}(|X^P(i)|) + b^P_{k-1}\right) \end{aligned} \quad (9)$$

where $X^E(i)$ and $X^P(i)$ represent the EEG spectrogram image and PPG spectrogram image, respectively, y^E_{k-1} and y^P_{k-1} are the outputs from the previous layers, and $W^E_{k-1}, W^P_{k-1}, b^E_{k-1}$, and b^P_{k-1} are the weights and biases applied to the inputs. The activation functions f^E_{k-1} and f^P_{k-1} are applied before fusion [28], [29]. The learned merge factors γ_1 and γ_2 control the contributions of each branch's features, facilitating an effective integration of complementary information from both modalities. This layer plays a crucial role in enabling accurate mental state classification by effectively balancing information from both the EEG and PPG spectrogram images. After fusion, the model passes the integrated features through LSTM and self-attention mechanisms to capture temporal dependencies and focus on the most relevant aspects of the mental state. Final classification is performed using a dense layer with a softmax activation function, predicting the mental state as one of the three classes: neutral, moderately negative, or severely negative.

The PhysiAttenConvLSTM combines CNNs, LSTMs, self-attention, and PIML for effective spatial–temporal feature extraction and multimodal fusion, offering a robust solution for real-time mental state classification with applications in mental health, human–computer interaction, and personalized healthcare.

F. Data Preparation and Training Strategies

In this study, EEG and PPG signals recorded by the edge device are segmented into 2-s intervals and filtered with fourth-order Butterworth bandpass filters (0.5–32 Hz for EEG and 0.4–5 Hz for PPG) to remove noise and preserve relevant activity. The filtered signals are standardized, then processed with HRST to generate spectrograms, which are normalized and resized to 28 × 28 pixels to ensure compatibility with lightweight neural network architectures commonly employed in resource-constrained edge computing scenarios. This resolution aligns with standard input sizes used in benchmark image classification tasks, such as the MNIST dataset, facilitating efficient inference with minimal computational overhead. The final dataset consisted of 53 370 spectrograms labeled as neutral, 49 832 as moderate negative, and 52 094 as severe negative. Fig. 5 shows the number of spectrogram samples for each class label: neutral, moderate negative, and severe negative.

To achieve a comprehensive evaluation of the model's performance, both ten-fold cross-validation (CV) and leave-one-out cross-validation (LOOCV) methods were employed. In the ten-fold CV approach, the dataset was randomly divided into ten equally sized folds, with each fold used once as the test set while the remaining folds were used for training; performance metrics were then averaged over all folds. To prevent temporal leakage, contiguous time segments from the same recording session were never split across folds; each segment was assigned entirely to either the training or test set. For the LOOCV method, a strict subject-independent protocol with temporal blocking was applied: in each iteration, all temporally contiguous data from one participant were held out entirely for testing, while the model was trained exclusively on data from the remaining participants. This ensured that no overlapping or temporally adjacent samples appeared in both training and testing, thereby eliminating overestimation due to autocorrelation and providing a robust assessment of the model's ability to generalize across individuals. Key performance metrics—accuracy (Acc), specificity (Spe), sensitivity (Sen), precision (Pre), and $F1$ score ($F1$)—are employed to comprehensively evaluate the classification capability of the model. All models are implemented in Python 3.11 utilizing the Keras framework with a TensorFlow backend. Training is conducted on a high-performance computing system equipped with an NVIDIA GeForce RTX 4070 GPU and an Intel Core i9-14900HX CPU.

Hyperparameter tuning plays a critical role in improving model training efficiency by identifying the most effective configuration settings. In this study, we utilize the tree-structured Parzen estimator (TPE) tuner, a sequential model-based optimization (SMBO) algorithm, to perform hyperparameter optimization. This process is implemented through Microsoft Research's AutoML toolkit, Neural Network Intelligence (NNI) [41], where we define the search space, select the optimization algorithm, and specify the evaluation metric. Table II provides a comprehensive overview of the specific hyperparameters used in our proposed model. The optimal training configuration comprises a batch size of 16, the Adam optimizer with a learning rate of 0.0001, and categorical cross-entropy as the loss function, trained over 250 epochs. The PIML module integrates learned merge factors (γ_1 = 0.7 and γ_2 = 0.3) to balance EEG and PPG features, as discussed in Section III-C.

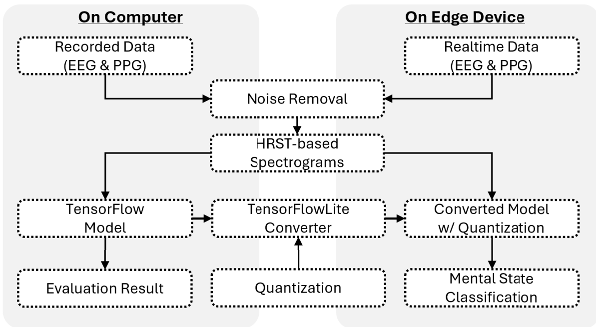

Fig. 6. On-edge processing pipeline for real-time mental state detection.

G. On-Edge Processing Pipeline

Our system executes the entire sequence of tasks—including multimodal signal acquisition (EEG and PPG), noise filtering, HRST-based spectrogram transformation, and mental state classification—directly on the edge device. This end-to-end workflow, illustrated in Fig. 6, enables real-time processing within a compact, wearable system, effectively reducing latency and data loss associated with traditional setups that rely on continuous data transmission to external computers. By eliminating such dependencies, the system achieves greater efficiency and reduced complexity. For model deployment on memory-constrained embedded hardware, we use TensorFlow Lite for Microcontrollers (TFLM), which allows TensorFlow-based neural networks to run on low-power microcontrollers. Trained models are converted into an optimized FlatBuffer (FB) format, storing 32-bit floating-point weights and compiled into C/C++ code. To further improve performance, we apply min–max quantization, reducing weight precision to 8-bit integers while preserving accuracy. This method computes a scale factor $s = (\max - \min/2^b - 1)$ and a zero point $z = \lfloor(-\min/s)\rceil$ to enabling efficient integer representation. Quantized values $q = \lfloor(x/s)\rceil + z$ are later dequantized for inference using $x' = s.(q - z)$. The resulting C/C++ FB code and TFLM runtime are stored in device memory. Model updates require only replacing FB weights, which can be transmitted wirelessly via Wi-Fi from the IoMT platform [42].

III. Results and Discussion

A. Signal Quality Evaluation and Artifact Removal

The quality of EEG signals acquired from BTE electrode placements on the proposed edge device was assessed using the classical "Berger effect" test [43]. This method evaluates the presence of increased alpha-band (8–14 Hz) activity during eyes-closed conditions, thereby serving as a validation of signal integrity. Participants alternated between eyes-open and eyes-closed states at two-second intervals. The application of the HRST method to the EEG recordings revealed a marked enhancement in alpha power (~10 Hz) during eyes-closed periods, with minimal changes during eyes-open periods, as illustrated in Fig. 7. In addition to spectral analysis, Fig. 7 presents the representative subfigures demonstrating both qualified and unqualified EEG signals following the application of a fourth-order Butterworth bandpass filter (0.5–32 Hz).

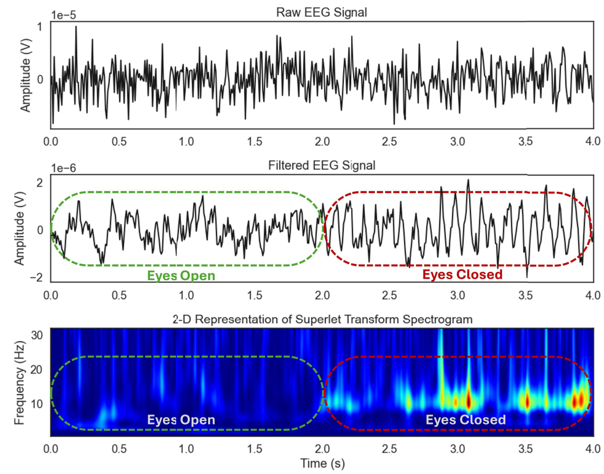

Fig. 7. Evaluation of BTE EEG signal quality using the "Berger effect" under eyes-closed and eyes-open conditions.

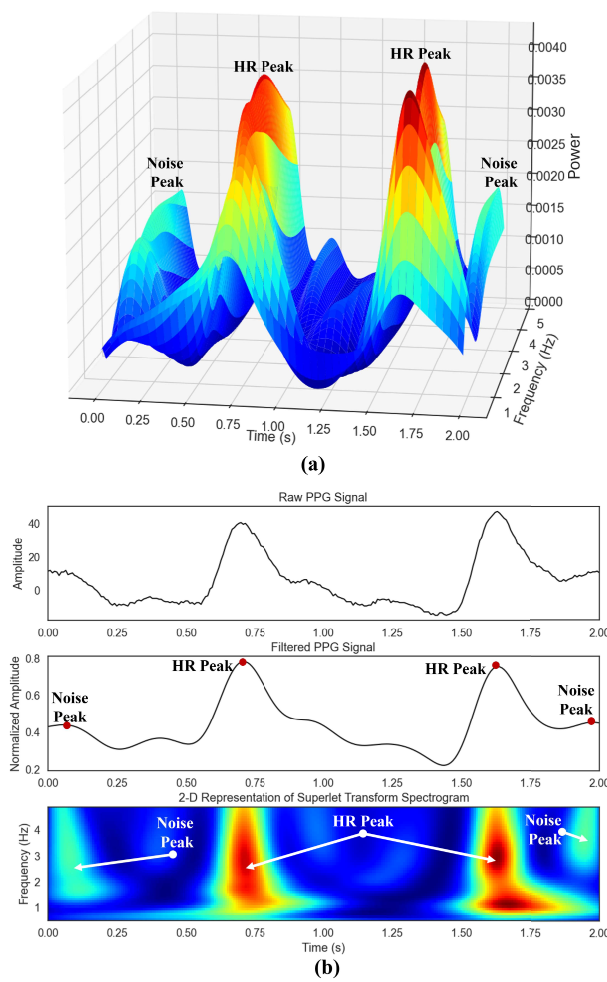

Fig. 8. (a) Separation of motion artifacts and dominant HR peak in in-ear PPG using HRST. (b) 2-D HRST spectrogram representation.

In the case of PPG signal analysis, a major challenge lies in the spectral overlap between heart rate (HR) components and motion artifacts, both occupying the 0.4–5-Hz frequency

TABLE III
EVALUATION OF NOISE REMOVAL USING RRMSE IN TEMPORAL AND SPECTRAL DOMAINS

Signal	$RRMSE_{temporal}$	$RRMSE_{spectral}$
In-ear PPG	0.284	0.263
BTE EEG	0.343	0.310

range. This overlap can result in erroneous HR estimations, thereby compromising mental state recognition accuracy. To address this issue, the HRST method is employed to differentiate HR-specific spectral peaks from motion artifacts, enhancing both HR estimation and downstream classification reliability. Importantly, the HRST approach operates without auxiliary sensors, such as accelerometers, thereby maintaining a compact and energy-efficient system design [44]. Fig. 8(a) displays a 3-D spectrogram of a PPG signal processed using HRST, highlighting the separation between HR and motion components by amplitude intensity, while Fig. 8(b) provides a 2-D projection to further illustrate the spectral distinction.

To quantitatively assess the effectiveness of the proposed signal processing approach, two objective performance metrics were utilized: the relative root mean square error (RRMSE) in the temporal domain, defined as $\mathrm{RRMSE}_{\mathrm{temporal}} = (\mathrm{RMS}(X - \tilde{X})/\mathrm{RMS}(X))$, and the RRMSE in the spectral domain, calculated using power spectral density (PSD) as $\mathrm{RRMSE}_{\mathrm{spectral}} = (\mathrm{RMS}(\mathrm{PSD}(X) - \mathrm{PSD}(\tilde{X}))/\mathrm{RMS}(\mathrm{PSD}(X)))$. In both cases, X denotes the clean EEG or PPG signal obtained via ICLabel—a robust ICA-based toolbox—combined with manual visual inspection [45], [46], while $\tilde{X}$ represents the signal after artifact removal. Although a computationally efficient fourth-order Butterworth bandpass filter was used to maintain real-time suitability for edge deployment, the results presented in Table III demonstrate that this method achieves high-quality signal restoration, as reflected in both temporal and spectral RRMSE metrics.

B. Mental State Classification Results

This study proposes a multimodal mental state recognition system that integrates EEG signals recorded from BTE electrode placements and PPG signals obtained from the in-ear canal. To ensure high signal quality and enable real-time on-device processing, raw EEG and PPG signals are first filtered using a fourth-order Butterworth bandpass filter—0.5–32 Hz for EEG to retain major cognitive-related brainwave bands and suppress noise [25], [57], [58] and 0.4–5 Hz for PPG to capture HR-related information while reducing baseline drift and motion artifacts [44], [59], [60], [61]. Following filtering, the signals are segmented into 2-s intervals and normalized. To convert time-series signals into a rich time–frequency representation, we employ the HRST, which enhances the spectral resolution without the need for additional sensors. The spectrogram pairs are utilized as dual-branch inputs to the proposed PhysiAttenConvLSTM model, which classifies mental states into three categories: neutral, moderate negative, and severe negative. The model's performance is evaluated using both ten-fold CV and LOOCV, following the training strategies and hyperparameter settings described in Section II-F. To assess the effectiveness of the proposed architecture, three baseline models are employed for comparison: 1) a CNN-based model comprising only the convolutional and fully connected layers of the PhysiAttenConvLSTM; 2) a ConvLSTM model that excludes the self-attention module, retaining only the convolutional, LSTM, and fully connected components; and 3) a 1-D ConvLSTM model that maintains the same architecture as the ConvLSTM baseline but replaces the 2-D convolutional layers with 1-D convolutional layers. As presented in Table IV, the proposed PhysiAttenConvLSTM model consistently outperforms all baseline models across key evaluation metrics, demonstrating its superior capability to extract and integrate discriminative features from EEG and PPG spectrograms for accurate mental state classification.

TABLE IV
CLASSIFICATION PERFORMANCE OF THE PROPOSED PHYSIATTENCONVLSTM AND TWO BASELINE MODELS UNDER LOOCV AND TEN-FOLD CV

Strategy	Model	Evaluation Indicators				
		Acc	Spe	Sen	Pre	F1
10-Fold CV	CNN	0.8655 ± 0.0224	0.8642 ± 0.0139	0.8613 ± 0.0221	0.8631 ± 0.0178	0.8592 ± 0.0127
	ConvLSTM	0.9192 ± 0.0302	0.9199 ± 0.0209	0.9231 ± 0.0305	0.9201 ± 0.0231	0.9189 ± 0.1307
	1D-ConvLSTM	0.8824 ± 0.0272	0.8656 ± 0.0391	0.8692 ± 0.0372	0.8672 ± 0.0318	0.8752 ± 0.0317
	PhysiAtten ConvLSTM	0.9505 ± 0.0128	0.9501 ± 0.0203	0.9499 ± 0.0139	0.9501 ± 0.0279	0.9496 ± 0.0152
LOOCV	CNN	0.8421 ± 0.0288	0.8453 ± 0.0482	0.8548 ± 0.0237	0.8492 ± 0.0306	0.8434 ± 0.0452
	ConvLSTM	0.8913 ± 0.0346	0.8891 ± 0.0225	0.8878 ± 0.0342	0.8889 ± 0.0429	0.8819 ± 0.0603
	1D-ConvLSTM	0.8534 ± 0.0262	0.8523 ± 0.0142	0.8527 ± 0.0215	0.8483 ± 0.0442	0.8548 ± 0.0534
	PhysiAtten ConvLSTM	0.9207 ± 0.0124	0.9193 ± 0.0179	0.9181 ± 0.0175	0.9205 ± 0.0152	0.9198 ± 0.0122

The results confirm that the proposed PhysiAttenConvLSTM model—which integrates convolutional layers, an LSTM with self-attention, and a physics-informed merge layer—consistently outperforms the baseline models in classifying mental states. Across both ten-fold CV and LOOCV, the model demonstrates superior performance in terms of accuracy, specificity, sensitivity, precision, and $F1$ score. Under ten-fold CV, the proposed model achieves an accuracy of 0.9505 ± 0.0128, outperforming the ConvLSTM (0.9192 ± 0.0302), 1D-ConvLSTM (0.8824 ± 0.0272), and CNN (0.8655 ± 0.0224) baselines. Similarly, in LOOCV, the model attains an accuracy of 0.9207 ± 0.0124, exceeding the performance of ConvLSTM (0.8913 ± 0.0346), 1D-ConvLSTM (0.8534 ± 0.0262), and CNN (0.8421 ± 0.0288). The slight reduction in LOOCV accuracy is attributed to intersubject variability, which poses challenges in individual-specific evaluations. Overall, these findings underscore the robustness and generalizability of the proposed model for real-world mental state classification using multimodal physiological data. Fig. 9(a) illustrates the performance comparison across folds under ten-fold CV, while Fig. 9(b) presents the performance comparison by subject (23 subjects) under

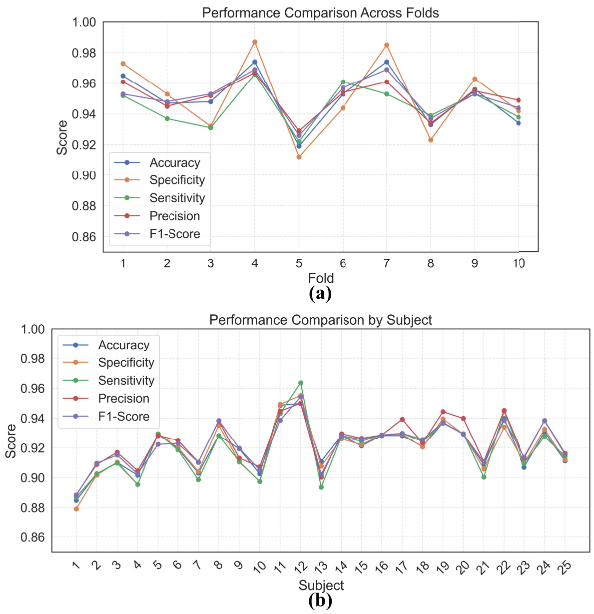

Fig. 9. Performance comparison (a) across folds using ten-fold CV and (b) by subject (25 subjects) using LOOCV.

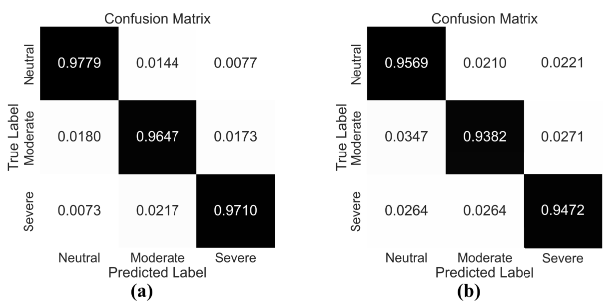

Fig. 10. Confusion matrices for (a) ten-fold CV (Fold 4) and (b) LOOCV (Subject 12).

LOOCV for the proposed PhysiAttenConvLSTM model in mental state classification. As shown in Fig. 9(a), under ten-fold CV, model performance varies across folds, achieving the highest scores across all five key evaluation metrics in Fold 4 and the lowest in Fold 5. Similarly, in Fig. 9(b), LOOCV results indicate performance variation across individual subjects, where all data from one participant are held out for testing. The best performance is observed for Subject 12, while the lowest is recorded for Subject 1. These variations reflect the inherent interfold and intersubject differences in physiological data, further emphasizing the model's adaptability and generalization capability. Fig. 10(a) and (b) displays the confusion matrices for the ten-fold CV and LOOCV settings, respectively.

We evaluated the performance of our multimodal negative state recognition system, which combines EEG signals from BTE locations with PPG signals from the in-ear canal. To assess the contribution of each modality, ablation experiments were conducted by isolating the EEG and PPG branches within the PhysiAttenConvLSTM model. Results show that the dual-branch approach significantly outperforms single-modality

TABLE V

Performance Comparison of the Proposed PhysiAttenConvLSTM With Three Input Configurations: EEG Only, PPG Only, and Combined (EEG and PPG)

Strategy	Input	Evaluation Indicators				
		Acc	**Spe**	**Sen**	**Pre**	**F1**
10-Fold CV	PPG	0.8992 ± 0.0201	0.8901 ± 0.0193	0.8877 ± 0.0218	0.8983 ± 0.0152	0.8876 ± 0.0228
	EEG	0.9135 ± 0.0142	0.9135 ± 0.0146	0.9131 ± 0.0202	0.9073 ± 0.0224	0.9153 ± 0.0213
	Both	0.9505 ± 0.0128	0.9501 ± 0.0203	0.9499 ± 0.0139	0.9501 ± 0.0279	0.9496 ± 0.0152
LOOCV	PPG	0.8602 ± 0.0149	0.8609 ± 0.0117	0.8597 ± 0.0143	0.8644 ± 0.0125	0.8598 ± 0.0215
	EEG	0.8883 ± 0.0168	0.8892 ± 0.0116	0.8814 ± 0.0217	0.8759 ± 0.0149	0.8791 ± 0.0223
	Both	0.9207 ± 0.0124	0.9193 ± 0.0179	0.9181 ± 0.0175	0.9205 ± 0.0152	0.9198 ± 0.0122

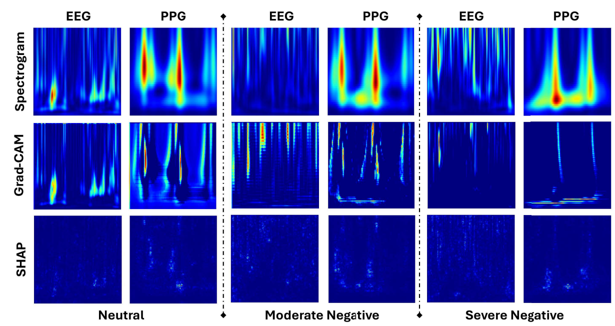

Fig. 11. Heatmaps showing Grad-CAM and SHAP feature attribution on EEG and PPG spectrograms for neutral, moderately negative, and severely negative mental states.

configurations in mental state classification (Table V). Among unimodal inputs, EEG consistently outperformed PPG across all evaluation metrics. Specifically, under LOOCV, EEG achieved an accuracy of 0.8883 ± 0.0168, while PPG reached 0.8602 ± 0.0149. In ten-fold CV, EEG attained 0.9135 ± 0.0142, compared to 0.8992 ± 0.0201 for PPG. These results highlight the central role of EEG, which captures direct neural correlates of mental processing, offering superior discriminatory power. While PPG reflects peripheral physiological responses, its link to mental states is more indirect. Combining EEG and PPG modalities enhances classification accuracy and robustness by leveraging both cognitive and physiological signals.

To elucidate the decision-making process of the proposed PhysiAttenConvLSTM model, we employed a two-pronged interpretability strategy combining gradient-weighted class activation mapping (Grad-CAM) [47] and SHapley Additive exPlanations (SHAP) [62] analysis. Grad-CAM was used to generate color-coded heatmaps over spectrogram representations derived from BTE-EEG and in-ear PPG signals, identifying spatial regions most influential to the model's predictions. In these heatmaps, areas highlighted in red correspond to features with strong positive contributions

TABLE VI
CLASSIFICATION PERFORMANCE COMPARISON WITH DIFFERENT PIML-LEARNED MERGE FACTORS

Trial	γ_1	γ_2	ACC
1	0.1	0.9	0.8404
2	0.2	0.8	0.8391
3	0.3	0.7	0.8543
4	0.4	0.6	0.8742
5	0.5	0.5	0.9201
6	0.6	0.4	0.9394
7	0.7	0.3	0.9505
8	0.8	0.2	0.9223
9	0.9	0.1	0.9154

TABLE VII
DURATION (ms), MCU CYCLES (kCYCLES), AND ENERGY COST (μJ) FOR EACH ON-EDGE PROCESSING TASK

On-Edge Processing Tasks		Duration		Energy Cost (μJ)	ACC
		in time (ms)	in MCU cycles (kCycles)		
Noise Filtering		10.31	824.82	165.6	---
HRST-based Spectrogram Conversion		712.27	56978.43	11306.3	---
Model Execution	Model w/o Quantization	59.67	4767.72	904.8	0.9207
	Model w/ Quantization	51.84	4285.41	786.3	0.9112
Post-Processing		1.14	105.22	22.95	---

to classification, thereby revealing modality-specific focus regions for each mental state. As shown in Fig. 11, these activation patterns are class-dependent, with EEG heatmaps emphasizing high-frequency beta activity in severe stress cases, while PPG heatmaps highlight low-frequency HRV-related patterns. To complement this spatial insight, we applied SHAP to quantify the contribution of individual spectrogram features across modalities, enabling a fine-grained analysis of the relative importance of EEG versus PPG in decision-making. The SHAP value distributions revealed that EEG features predominantly drive the differentiation between moderate and severe negative states, whereas PPG features contribute more to distinguishing neutral from negative states. By combining Grad-CAM's localized visualization with SHAP's feature attribution, we provide a more comprehensive interpretability framework that enhances clinical trust in the model's predictions.

C. Impact of PIML-Learned Merge Factors on Classification Performance

The PIML integrates features from dual CNN branches processing HRST-based EEG and PPG spectrograms using a weighted addition mechanism, allowing the model to dynamically balance each modality's contribution and enhance mental state classification performance. To optimize this balance, a grid search is performed, constraining the merge factors (γ_1 for EEG and γ_2 for PPG) to sum to 1. Results in Table VI demonstrate that the best performance (accuracy = 0.9505) is achieved when $\gamma_1 = 0.7$ and $\gamma_2 = 0.3$, confirming that EEG contributes more critical information than PPG for mental state recognition. Conversely, the poorest performance (accuracy = 0.8391) occurs when $\gamma_1 = 0.2$ and $\gamma_2 = 0.8$. Notably, increasing the EEG weight to 0.9 ($\gamma_1 = 0.9$) still yields better results than heavily weighting PPG ($\gamma_2 = 0.9$), emphasizing the superior discriminative power of BTE EEG. While both signals provide valuable physiological information, EEG's direct link to neural activity makes it more effective in capturing mental states. These findings highlight the importance of modality weighting and the need for balanced integration of multimodal signals to maximize classification accuracy.

D. On-Edge Deployment

Building on prior analysis, the proposed PhysiAttenConvLSTM model demonstrates strong performance in mental state classification by leveraging HRST-based spectrogram pairs of EEG and PPG signals as dual-branch inputs. To validate its real-world applicability, we implemented an end-to-end edge deployment, processing the entire pipeline—from signal acquisition and preprocessing to spectrogram generation and classification—directly on the edge device without relying on PCs or cloud servers. An LOOCV strategy enhances model generalization, with classification outputs postprocessed and transmitted to an IoMT cloud platform for storage and analysis. To ensure efficient on-device deployment in memory-constrained environments, we applied optimization techniques, including quantization, to reduce model size and latency while maintaining accuracy. Originally developed in TensorFlow, the model was adapted for edge deployment using TensorFlow Lite, which offers optimizations and an interpreter suited for low-power devices, enabling real-time inference for continuous mental state monitoring in wearable health applications.

Table VII summarizes the task-specific performance metrics of the proposed edge-based system, which operates entirely on-device. Key parameters include execution time (ms), MCU cycles, energy consumption (μJ), and model accuracy before and after quantization. Results show that while the accuracy of both the original TensorFlow and quantized TensorFlow Lite models is comparable, a slight accuracy reduction of 0.95% occurs—dropping from 92.07% to 91.12%—due to the shift from 32-bit floating-point to 8-bit integer weights, which reduces precision. However, quantization significantly improves processing speed, reducing inference time from 59.67 to 51.84 ms. This enhancement is particularly beneficial for battery-operated embedded systems, where low power consumption and fast response are crucial. The quantized model accelerates inference and substantially lowers energy usage and computational overhead. To evaluate these improvements, we used the Power Profiler Kit 2 (PPK2) from Nordic Semiconductor along with the Power Profiler App (PPA) to measure on-device latency and energy consumption. Table VII presents the measured execution time, MCU cycles, and energy consumption across various processing stages, highlighting the quantized model's suitability for efficient, real-time mental state classification on edge devices.

TABLE VIII
PROPOSED MODEL SIZE IN TERMS OF RAM (kB) AND FLASH (kB) USING TFLM

Optimization	Model Size (kB)	Memory (kB)		ACC
		FLASH	RAM	
Non-quantized Model	102.83	105.834	12.835	0.9207
Quantized Model	26.19	53.184	9.875	0.9112

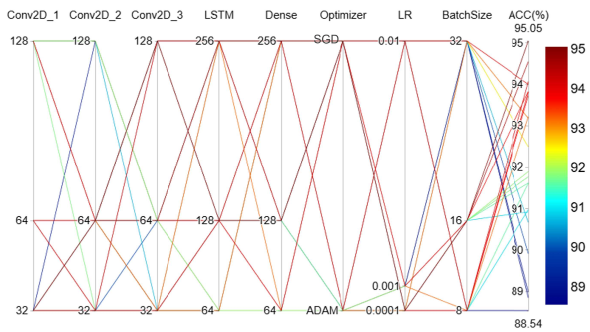

Fig. 12. Hyperparameter tuning results for the proposed PhysAttenConvLSTM model.

To further assess the system's power efficiency, the edge device was tested with a 3.7-V 500-mAh battery under continuous operation without sleep mode. The device exhibited an average power consumption of approximately 85.193 mW, resulting in an estimated battery life of 21.71 h. Table VIII compares the model's memory footprint before and after quantization using TFLM. The quantization process significantly reduces memory usage, with flash memory decreasing from 105.834 to 53.184 kB, RAM from 12.835 to 9.875 kB, and model weights from 102.83 to 26.19 kB, achieving a fourfold reduction.

E. Ablation Analysis

We employed the TPE, a Bayesian optimization method, through the NNI framework to efficiently explore the parameter space (refer to Section II-F). The search space included the number of filters in Conv2D_1, Conv2D_2, and Conv2D_3 layers, the number of units in the LSTM and dense layers, as well as the choice of optimizer, learning rate, and batch size. Among the configurations tested, the setting with Conv2D_1 = 32, Conv2D_2 = 64, Conv2D_3 = 128, LSTM = 128, dense = 128, optimizer = ADAM, learning rate = 0.0001, and batch size = 16 achieved the highest classification accuracy of 95.05%, as shown in Fig. 12. The second-best result (94.55%) was obtained using a configuration with Conv2D_1 = 32, Conv2D_2 = 64, Conv2D_3 = 64, LSTM = 256, dense = 256, optimizer = ADAM, learning rate = 0.0001, and batch size = 16. In contrast, the lowest accuracy (88.54%) was recorded with Conv2D_1 = 128, Conv2D_2 = 128, Conv2D_3 = 64, LSTM = 64, dense = 64, optimizer = SGD, learning rate = 0.01, and batch size = 8. These results highlight the substantial influence of hyperparameter configurations on model performance.

F. Comparison With Previous State-of-the-Art Studies

Numerous studies have investigated the use of biosignals, particularly EEG and PPG, for mental health monitoring. However, most of these studies rely on scalp EEG and PPG acquired from the wrist or fingers. Traditional configurations like these present several limitations for continuous, real-world applications. Scalp EEG systems are often bulky and obtrusive, requiring conductive gel, multiple electrodes, and time-consuming setup. Meanwhile, wrist-based PPG is susceptible to motion artifacts and may lose accuracy during physical activity. For example, Li et al. [12] proposed a graph-based multitask self-supervised learning approach for EEG-based mental state recognition using a 62-channel scalp EEG ESI NeuroScan system, achieving an accuracy of 86.52% across three emotional classes, with emotions induced by film clips. Similarly, Song et al. [48] developed an emotion recognition model based on dynamic graph CNNs using 16-channel scalp EEG from the Emotiv EPOC, achieving classification accuracies of 86.23%, 84.54%, and 85.02% for valence, arousal, and dominance, respectively. In another study, Kim et al. [49] introduced a dual-function, integrated emotion-based music classification system that utilized features extracted from wrist-based PPG and GSR signals using the Biosemi ActiveTwo system, combined with a convolutional bidirectional gated recurrent unit (Bi-GRU) network, achieving classification accuracies of 92.41% for valence and 90.27% for arousal on the DEAP database.

To address the limitations of multichannel EEG systems and the susceptibility of wrist- or finger-based PPG signals to motion artifacts, several researchers have aimed to simplify the sensing setup by reducing the number of signal acquisition sites. For example, Yu et al. [50] proposed using single-channel scalp EEG from the NeuroSky TGAM combined with digital twin technology through a three-branch spatial–temporal feature extraction network (TB-SFENet) for emotion recognition, achieving an accuracy of 82.50% across three emotional categories (positive, negative, and neutral) using music and video clips as emotional stimuli. Bird et al. [63] proposed mental emotional sentiment classification with four scalp EEG channels from the MUSE EEG headband, achieving an accuracy of 94.89% across three emotional categories (positive, negative, and neutral) using film clips as emotional stimuli. In another approach, Zhao et al. [51] utilized chest-based PPG and ECG signals from Respiban Professional sensors and the Empatica E4 wrist-worn device instead of traditional wrist- or finger-based PPG, employing a contrastive autoencoder and achieving 90% accuracy for two emotional categories (positive and negative) on the WESAD dataset, a multimodal database for wearable stress and affect detection.

Given the previously discussed limitations of scalp EEG and wrist-based PPG, ear-centric physiological sensing—particularly ear EEG and in-ear PPG—has emerged as a promising alternative, offering an optimal balance between signal quality, user comfort, and practical

TABLE IX
COMPARISON OF KEY FEATURES BETWEEN OUR PROPOSED SYSTEM AND PREVIOUS STATE-OF-THE-ART MENTAL STATE MONITORING SYSTEMS

Study /Published Date	No. of Subjects	Dataset (Device)	Stimuli	Measurement Location/Modalities	Methodology	Results	Processing
[48] 2018	23	DREAMER (Emotiv EPOC)	Film clips	Scalp EEG (16 channels)	EEG Emotion Recognition via Dynamical Graph Convolutional Neural Networks	Accuracy: 86.23% for valence, 84.54% for arousal, and 85.02% for dominance	On PC
[63] 2019	2	MUSE EEG headband	Film clips	Scalp EEG (4 channels)	Mental Emotional Sentiment Classification with an EEG-based Brain-Machine Interface	Accuracy: 94.89% for 3 emotional classes (positive, neutral, and negative)	On PC
[49] 2021	18	DEAP (Biosemi ActiveTwo system)	Music videos	GSR and PPG	A Dual-Function Integrated Emotion-Based Music Classification System Using Features from Physiological Signals	Accuracy: 92.41% for valence classification and 90.27% for arousal classification	On PC
[12] 2022	12	SEED (ESI NeuroScan System)	Film clips	Scalp EEG (62 channels)	GMSS: A Graph-Based Multi-Task Self-Supervised Learning Approach for EEG Emotion Recognition	Accuracy: 86.52% for 3 emotional classes (positive, negative, neutral)	On PC
[52] 2023	21	A custom-designed device	Auditory stimulus	Around-ear EEG	Decoding Auditory-Evoked Responses in Affective States Using a Wearable Around-Ear EEG System	Accuracy: 67.09% for valence classification and 66.61% for arousal classification	On PC
[50] 2024	10	NeuroSky TGAM	Music & videos	Scalp EEG (1 channel)	An Intelligent Wearable System for Motion and Emotion Recognition Using Digital Twin Technology	Accuracy: 82.50% for 3 emotional classes (positive, negative, neutral)	On PC
[18] 2024	14	A custom-designed device	Emotional videos	BTE EEG (3 channels)	A Deep Learning-Based Wearable Ear-EEG Emotion Recognition System with a Superlets-Based Signal-to-Image Conversion Framework	Accuracy: 92.41% for binary classification (negative and positive)	On PC
[51] 2025	15	WESAD (Respiban Professional sensor & Empatica E4 wrist-worn device)	Trier Social Stress Test (TSST)	ECG and PPG	Multi-source Signal Fusion with Contrastive Auto Encoder for Emotion Classification	Accuracy: 90% for 2 mental state classification (positive and negative)	On PC
Ours	**25**	**Multimodal ear-centric edge device**	**Social media**	**BTE EEG and In-ear PPG**	**End-to-End On-Edge Physics-Informed Neural Networks for Detecting Negative Mental States in IoMT**	**Accuracy: 95.05% using 10-fold CV for 3-level mental state classification. Accuracy: 92.07% using LOOCV for 3-level mental state classification.**	**On Edge**

applicability. Notable studies in this domain include Mai et al. [18], who employed BTE EEG recordings from three electrodes combined with a modified vision transformer (ViT) model for emotion recognition, achieving an accuracy of 92.41% for two emotional classes (negative and positive) using an emotional video dataset. Similarly, Choi et al. [52] proposed a method to decode auditory-evoked responses associated with affective states using a wearable around-the-ear EEG system and a multilayer extreme learning machine, attaining accuracies of 67.09% for valence and 66.61% for arousal under auditory-based emotional stimuli. Table IX provides a comprehensive comparison between the proposed system and previous state-of-the-art mental state monitoring approaches.

Although previous studies have demonstrated high performance in mental state monitoring (emotion recognition) using EEG and PPG signals, they often rely on external computing resources—such as smartphones, PCs, cloud servers, or edge gateways—for signal preprocessing, feature extraction, and classification. While these architectures benefit from powerful computation, they face limitations, including increased latency, higher energy consumption, and reliance on stable connectivity. These issues are particularly problematic for continuous, real-time mental state monitoring in mobile or uncontrolled environments, where low latency, extended battery life, and user privacy are crucial. For instance, Gangadharan et al. [53] employed the Muse-2 device featuring four temporal EEG channels, with a power consumption of 214 mW and an operational duration of up to 5 h on a 290-mAh battery. Pham et al. [54] developed a BTE EEG system comprising eight channels, which draws 241.3 mW from a 600-mAh battery and delivers approximately 9.2 h of operation. In a related effort, Budidha et al. [55] proposed a modular, multichannel PPG monitoring system with a power consumption of 1.5 W, powered by a 176-mAh battery, and capable of operating for approximately 4 h. Similarly, Chacon et al. [56] introduced a wearable pulse oximeter with wireless communication capabilities, reporting a total system power consumption in the range of 450–530 mW.

In this study, we propose a novel ear-centric wearable system combining BTE EEG and in-ear PPG, integrated with PINNs deployed entirely on-device for detecting negative mental states. By embedding physiological knowledge into the model, the system enhances robustness and interpretability while preserving low-power performance. Our edge device, powered by a 3.7-V, 500-mAh battery, consumes

TABLE X
WEARABILITY AND USABILITY EVALUATION DURING ONE-WEEK SOCIAL MEDIA USAGE PILOT STUDY

Subject	Daily Social Media Usage (h/day)	Compliance Rate (%)	Comfort Rating (1–5)	Slippage/ Displacement Events	Sensor Drift Observed
1	3.2	95	5	0	No
2	2.8	90	4	1	No
3	4.0	98	5	0	No
4	3.5	92	4	0	No
5	2.6	88	4	0	No
6	3.8	96	5	0	No
7	2.9	89	4	0	No
8	3.3	93	5	0	No
9	2.7	87	4	0	No
10	3.6	94	5	0	No

only 85.193 mW during continuous operation, offering an extended battery life of approximately 21.71 h—enabling real-time, private, and energy-efficient mental health monitoring.

To evaluate the effectiveness of the proposed system in detecting negative mental states, we conducted a study with 25 participants, each undergoing five experimental sessions. Model performance was assessed using five evaluation metrics under ten-fold CV and LOOCV. The LOOCV strategy provided subject-independent validation, training the model on data from all but one participant, offering insights into the system's robustness and generalizability. Despite promising results, further studies are needed to confirm the system's applicability in uncontrolled, real-world settings. Future research will focus on expanding the dataset with more participants, increasing trial frequency for better training depth, and evaluating the system across diverse demographic and physiological profiles to ensure reliable performance across various user populations.

G. Wearability and Usability Evaluation

To evaluate the device's long-term wearability and usability under realistic daily conditions, we conducted a one-week pilot study in a social media usage scenario—a common, cognitively engaging activity known to elicit diverse emotional and physiological responses. Ten participants, randomly selected from the original cohort of 25, were instructed to wear the device whenever accessing social media platforms during their normal routines. As summarized in Table X, the average daily usage time was 3.24 $\pm$ 0.48 h/day, with a mean compliance rate of 92.2% $\pm$ 3.6% and a high comfort rating (4.5 $\pm$ 0.3 on a 1–5 scale). Minor slippage events were reported by only one subject, and no noticeable sensor drift occurred during the trial. These results indicate that the proposed ear-worn design can be comfortably and reliably used for prolonged, naturally occurring activities involving emotional variability, while also highlighting the need for larger scale, long-duration studies (e.g., continuous 24-h wear or multiweek trials) to fully validate real-world feasibility.

IV. CONCLUSION

This study presents a novel, fully integrated ear-centric system for real-time mental state monitoring that addresses key limitations of conventional approaches in terms of mobility, latency, and energy efficiency. By combining multimodal biosignal acquisition from BTE EEG and in-ear PPG sensors with high-resolution spectrogram generation via the HRST and a lightweight, edge-deployable DL architecture, the proposed system demonstrates strong classification performance with minimal computational cost. The incorporation of a physics-informed merge layer enhances both robustness and interpretability, while the system's IoMT compatibility supports scalability and remote accessibility. Empirical evaluation on data collected from 25 participants during structured social media-based mental state induction tasks yielded promising results, with classification accuracies of 92.07% (LOOCV) and 95.05% (ten-fold CV), as well as high sensitivity, specificity, precision, and $F1$ scores. These findings underscore the system's potential for practical deployment in affective computing, mobile health monitoring, and personalized mental wellness applications.

Future research will aim to improve model generalizability by expanding the dataset to include more heterogeneous populations and incorporating additional physiological modalities. Although the current experimental protocol simulates realistic social media interactions, it remains limited by its controlled, task-based design. To enhance ecological validity, future studies should explore naturalistic scenarios involving unstructured daily activities, such as commuting, studying, or socializing, while accounting for factors, such as movement artifacts, ambient noise, device repositioning, and long-term physiological variation. We also acknowledge that confounding physiological states, including fatigue, drowsiness, and circadian rhythm variations, were not explicitly controlled in the current study and may influence EEG/PPG signals. Accordingly, future work will incorporate auxiliary data, such as actigraphy, sleep history, caffeine intake, and alertness measures, to disentangle stress-specific patterns from other physiological effects. Additionally, the potential inclusion of personal demographic information (e.g., age, gender, BMI) as input features will be explored to enhance model personalization and robustness without compromising computational efficiency. Finally, we note the limitation posed by the 0.4-Hz lower cutoff used in PPG signal filtering, which may attenuate components from atypically low HRs; future studies will investigate alternative filter configurations to better accommodate clinical populations or individuals with slower HR profiles.

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